

British Telecommunications plc.

## **SPECIFICATION LN435-N**

### **FOR**

### **WOOD POLES**

#### **SCOPE**

This specification details the requirements for Wood Poles for the support of BT Overhead Plant. For poles treated outside of the UK, additional requirements apply, and these are detailed in supplementary specification LN436, which should be read in conjunction with this document.

#### **CONTENTS**

|           |                                     |
|-----------|-------------------------------------|
| <b>1</b>  | <b>GENERAL NOTES</b>                |
| <b>2</b>  | <b>SPECIES</b>                      |
| <b>3</b>  | <b>SOURCES</b>                      |
| <b>4</b>  | <b>INSPECTION AND DEFECTS</b>       |
| <b>5</b>  | <b>CONVERSION</b>                   |
| <b>6</b>  | <b>STACKING</b>                     |
| <b>7</b>  | <b>DIMENSIONS</b>                   |
| <b>8</b>  | <b>MARKING AND DRILLING</b>         |
| <b>9</b>  | <b>KILN DRYING</b>                  |
| <b>10</b> | <b>PRESERVATION</b>                 |
| <b>11</b> | <b>GLOSARY OF TERMS</b>             |
| <b>12</b> | <b>BLEEDING SCALE</b>               |
| <b>13</b> | <b>PRODUCT APPROVAL</b>             |
| <b>14</b> | <b>REFERENCES</b>                   |
| <b>15</b> | <b>HISTORY</b>                      |
| <b>16</b> | <b>SPECIFICATION AUTHORITY (SA)</b> |

## **1 GENERAL**

### **1.1 General Notes**

All details, dimensions and instructions shown on any drawings, diagrams and specifications quoted in this specification shall be taken as forming part of this specification. The latest issues of the relative drawings, diagrams and specifications will be indicated in the invitation to tender. The contractor must ensure that the drawings and specifications used are in accordance with the detail appearing in the contract schedule.

### **1.2 Numbering**

The first issue of a specification carries a number without a suffix letter. When a specification is re-issued a suffix letter is added and for each subsequent issue the suffix letter is alphabetically advanced by one.

### **1.3 Amendment**

Except when a specification is completely rewritten, an amended paragraph is indicated by a star adjacent to the paragraph number and the particular portion amended is indicated by a vertical line in the margin.

### **1.4 Supplementary Specifications**

When a British Standard or other specification is quoted by number only, the latest issue of that specification is to be followed, but if, in addition to the number, a particular issue is quoted that issue is to be followed.

### **1.5 Patterns, Drawings and Specifications**

If there is any discrepancy between patterns and drawings, the drawings shall be followed, and if there is discrepancy between drawings and specification, the specification shall be followed. All discrepancies shall be reported to the SA.

### **1.6 Standard Patterns**

When standard patterns are employed, supplies shall be at least equal as regards any properties, qualities or dimensions (other than minor non-essential differences due to variations in trade practice and which do not affect serviceability) not defined in this specification, to the standard patterns.

## **1.7 Information**

If any further information in connection with this specification is required, application should be made to BT, Access Engineering & Innovation.

## **1.8 Specification Authority**

The Specification Authority (SA) is responsible for Product Approval. The reference for the BT Specification Authority is given at the end of the Product Specification.

## **2 SPECIES**

Poles shall only be from timbers of the following species, *Pinus sylvestris* (Scots Pine, European Redwood). Poles shall comply with all requirements of BS EN 14229:2010, unless detailed in this specification.

Poles from other species may be considered, but only after application to, and agreement in writing from the SA, prior to any supply.

## **3 SOURCES**

### **3.1 Geographical Areas**

Scots Pine/European Redwood shall be sourced from Nordic geographical areas north of Latitude 55°N only. For the purposes of this specification, Nordic is Sweden, Norway & Finland only.

Poles from home-grown sources are no longer permitted, from 01/01/2014.

Poles from other sources may be considered, but only after application to, and agreement in writing from the SA, prior to any supply.

### **3.2 Test Data**

Within 3 months of contract, the supplying company shall provide to the SA a report on pole strength test results, verified by an independent authority.

Due to the nature of poles and the supply chain, it is accepted that the same supplier may continue to supply poles on the same contract, or which has simply been renewed. In this case, it is expected that the maximum period for which test results and data is valid is 5 years. After this time, it will be necessary for incumbent suppliers to arrange for

repeat testing. The SA will give a minimum of 6 months' notice to suppliers for repeat testing.

The data to be provided will give the mean and standard deviations of the breaking loads and rupture stresses of poles tested to their breaking load. Raw test data shall be provided with the report. Data for a minimum of 50 poles shall be provided.

The report shall be certified by an independent authority/test house as being representative of the poles being supplied in terms of source areas, defects, species, preservation, shaping and sizes.

### **3.3 Environmental Factors**

Poles should be supplied from forests which:

- Are managed in conformity with national laws.
- Are specifically managed for commercial harvesting. Any use of "old growth" forests should be justified and its potential environmental effects documented by a formal impact assessment.

The raw material supplier should have a stated policy towards the management of its forested supply areas in terms of sustainability and minimising adverse environmental impacts judged over the whole tree cycle. This should address the issues of planting diversity, planting of non-indigenous species, wildlife protection, landscape value, use of chemicals and logging practices.

Logging of trees should be conducted in such a way so as to minimise adverse impacts on wildlife and soil/water regimes. With respect to these potential impacts, there should be a stated policy on the size of individual logging areas and post logging practices.

Submissions to BT in accordance with GS13 shall address the above areas specifically.

## **4 INSPECTION AND DEFECTS**

### **4.1 Inspection for Defects**

All poles shall be inspected after dressing and not more than one month before preservation.

### **4.2 Decay**

Poles shall be of sound wood, rot or punks are not permitted.

### 4.3 Knots

Decayed or loose knots are not permitted. Soft knots containing loose or soft fibres are permitted provided that the affected area does not exceed 15 mm diameter and the depth is less than 10 mm. Depth to be measured with a BT Pole Probe or a handled 6.7 mm diameter steel spike, the end 12 mm of which shall be shaped to a sharp point. Insertion force shall be  $12 \pm 1$  kg.

The term knot in this clause shall include knots and knot clusters.

At any cross section of the pole at  $90^\circ$  to the longitudinal axis of the pole, the sum of the surface diameters of any knots of which any visible part is present on the cross section shall not exceed 0.2 times the circumference at that point.

The number of knots of which any visible part is present on the cross section shall not exceed five.

For poles up to and including 13 m length and for the bottom two-thirds of poles over 13 m in length, no individual knot diameter measured at  $90^\circ$  to the longitudinal axis of the pole shall exceed 0.065 times the pole circumference at that cross section or 50 mm, whichever is the smaller.

For the top one-third of poles over 13 m in length no knot diameter measured at  $90^\circ$  to the longitudinal axis of the pole shall exceed 0.08 times the pole circumference at that cross section.

Depressions adjacent to knots in excess of 5 mm in depth are not permitted.

### 4.4 Insect Holes

In general, insect damage shall be avoided, and poles with active insect infestation at or following final inspection are not acceptable.

Poles containing holes greater than 1.5 mm diameter (shot holes or grub holes) are not acceptable.

Poles containing a number of pinholes (holes less than 1.5 mm diameter) are acceptable within the following limitations.

- Pinholes are not permitted within 2 m of the tip.
- The remaining surface area shall not contain more than 50 pinholes evenly spaced in any 300 mm section.
- The surface area of the pole shall not contain more than 200 pinholes in total.

#### 4.5 Damage

The following are not permitted on a dressed pole except as otherwise allowed in this section: -

Animal bites, bird damage, inbarks, mechanical damage, rind galls

Mechanical damage shall be limited to broken surface fibres whose depth must not exceed 5 mm and where the damage shall be removed by trimming.

Rind galls, animal bites, bird damage and inbarks with a depth of 5 mm or less may be removed as described in mechanical damage.

Removal of defects as allowed above is restricted to 2 occurrences on a pole, and they must not be on the same cross section so as to cause wasting of the pole.

Rind galls within 900 mm of the butts are allowed with the following restrictions on size:

-

Length 1 x Diameter of pole at mid-point of defect.

Width 0.25 x Diameter of pole at mid-point of defect.

Depth 0.25 x Diameter of pole at mid-point of defect.

Depressions in excess of 5 mm in depth are not permitted.

Compressed wood caused by open stacking shall be limited to no more than a total decrease in diameter of 5 mm measured across the pole.

#### 4.6 Sapstain

Blue stain in the sapwood shall not be considered a defect.

#### 4.7 Growth and Growth Rings

The pole shall have a 25 mm minimum depth of sapwood, around the full circumference of the pole, after dressing. The diameter of the area of Juvenile wood (the relative area enclosed by the 12<sup>th</sup> growth ring from the pith) shall not exceed 50% of the pole diameter when measured at the tip.

It is accepted that due to being cut as close to the ground line as possible, some poles will have Butt Flutes. Where such Butt Flutes exist, it may not be possible to verify a 25mm sapwood ring fully around the butt of the pole.

In these exceptions, it is permitted to verify the presence of a 25mm sapwood ring all around the pole by taking a core sample from the pole, directly above the butt flute, and no more than 600mm from the butt of the pole.

The growth rings should be closely pitched with *Pinus sylvestris* having a minimum of 8 rings per 25 mm.

#### 4.8 Fissures

Cracks across poles or across the grain shall not be permitted. Seasoning shakes (or checks) along the grain are expected and are not recognised as defects within the following limits. Shakes shall not have a depth greater than half the diameter at any point along the pole, when measured with a 0.2 mm feeler gauge. No shake shall be continuous from pole butt to tip.

#### 4.9 Ring and Star Shakes

The tip shall be free from ring or star shakes but one complete ring or one star shake is acceptable at the butt provided that no part extends to within 50 mm of the periphery.

#### 4.10 Slope of Grain

Spiral grain shall not exceed 1 in 10 measured over any 1 m along the pole.

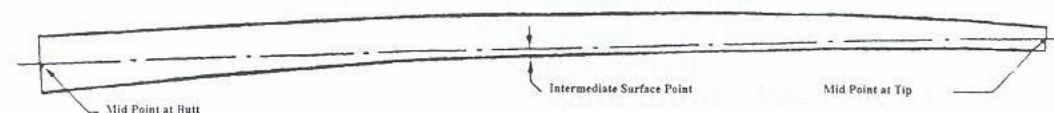
#### 4.11 Compression Wood

Compression wood is not permitted.

#### 4.12 Straightness

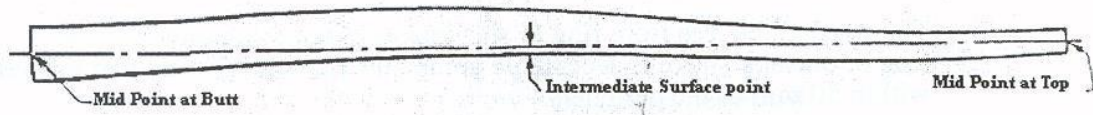
The pole should have a uniform taper. Short crooks, double sweeps, butt sweeps are not allowed. Butt flutes are allowed within the limitations on size (Section 7).

A single sweep is allowed to the extent that a straight line from the centre of the butt to the centre of the tip, shall not at any intermediate point pass nearer to the surface of the pole than one quarter of the diameter at the point in question. See Diagram 1, below.



**DIAGRAM 1 – MEASUREMENT OF SWEEP IN ONE PLANE AND ONE DIRECTION**

A reverse sweep where the sweep is in 2 directions and one plane is allowed, to the extent that a straight line from the centre of the butt to the centre of the tip shall not at any intermediate point pass nearer to the surface of the pole than one-third of the diameter at the point in question. See Diagram 2 below.



**DIAGRAM 2 – MEASUREMENT OF REVERSE SWEEP IN TWO DIRECTIONS AND ONE PLANE**

## **5 CONVERSION**

### **5.1 Felling**

Poles shall be felled between 1 November and 31 March of the following year.

### **5.2 Butts**

Poles shall be from trees sawn off as close to the ground as possible, the butt of the pole being the base of the tree. Where cutting or trimming is necessary to meet size limitations contained elsewhere in this specification it shall be kept to an absolute minimum. Poles shall not be provided from trees where a commercial sawlog has been taken from the lower section of the tree below the butt of the pole.

### **5.3 Trimming**

The outer and under bark should be removed as soon as possible after felling. Swelling of wood around the knots shall be completely removed.

The butt of the pole shall be sawn off at  $90^{\circ} \pm 10^{\circ}$  to the longitudinal axis of the pole.

There should be no abrupt change in the contour of the pole surface due to bark removal, trimming, shaving, dressing or any other process.

## **6 STACKING**

### **6.1 Removal from Forest**

The poles shall be removed from the forest within one month of being felled.

## 6.2 Storage

At the depot the poles shall be stacked clear of the ground, in open formation, at least 250mm above the ground. Any dunnage used shall be free from rot, and sufficient dunnage used to distribute the total mass without imposing undue strain on poles in the lower layers.

The surface of the ground shall be well drained, kept free from bark, shavings, grass and weeds.

## 7 DIMENSIONS

### 7.1 Sizes and Weights

After felling, conversion and final dressing (i.e. machine dressing at Pole Suppliers Depot), the pole shall be in accordance with the dimensions for its class as shown in Tables 1, 2 and 3. The diameter at the top shall be measured at the bolt hole as detailed in Drawing CN 206. The tolerance on length shall be – 50/+ 150 mm.

Butt flutes shall not cause the minimum diameter at the butt to be less than that required at 1.5 m. Maximum butt sizes shall be such that the butt of the pole will fit into a circular hole of diameter 300 mm for light poles, 350 mm for medium poles and 400 mm for stout poles.

On delivery poles shall not exceed the weights shown in Tables 1,2 & 3.

**Table 1**

| SIZES AND WEIGHTS OF LIGHT POLES |                     |                    |     |                                           |                                |                   |
|----------------------------------|---------------------|--------------------|-----|-------------------------------------------|--------------------------------|-------------------|
| LENGTH                           | ITEM<br>CODE<br>Nos | DIAMETER AT<br>TOP |     | MIN. DIAMETER<br>AT 1.5m FROM<br>BUTT END | MAXIMUM<br>WEIGHT IMP.<br>RED. | NOMINAL<br>VOLUME |
|                                  |                     | MIN                | MAX |                                           |                                |                   |
|                                  |                     | mm                 | mm  | mm                                        | kg                             | m <sup>3</sup>    |
| 6                                | 013041              | 125                | 175 | 150                                       | 112                            | 0.110             |
| 7                                | 013042              | 125                | 175 | 160                                       | 134                            | 0.129             |
| 8                                | 013043              | 125                | 175 | 170                                       | 165                            | 0.199             |
| 9                                | 013044              | 125                | 175 | 180                                       | 203                            | 0.225             |
| 10                               | 013045              | 125                | 175 | 185                                       | 245                            | 0.270             |
| 11                               | 013046              | 125                | 175 | 195                                       | 276                            | 0.320             |
| 13                               | 013047              | 130                | 175 | 210                                       | 391                            | 0.430             |

Table 2

| SIZES AND WEIGHTS OF MEDIUM POLES |                     |                    |     |                                           |                                |                   |
|-----------------------------------|---------------------|--------------------|-----|-------------------------------------------|--------------------------------|-------------------|
| LENGTH                            | ITEM<br>CODE<br>Nos | DIAMETER AT<br>TOP |     | MIN. DIAMETER<br>AT 1.5m FROM<br>BUTT END | MAXIMUM<br>WEIGHT IMP.<br>RED. | NOMINAL<br>VOLUME |
|                                   |                     | MIN                | MAX |                                           |                                |                   |
|                                   |                     | mm                 | mm  | mm                                        | kg                             | m <sup>3</sup>    |
| 7                                 | 013048              | 140                | 175 | 200                                       | 194                            | 0.210             |
| 8                                 | 013049              | 145                | 180 | 210                                       | 230                            | 0.240             |
| 9                                 | 013050              | 150                | 180 | 220                                       | 275                            | 0.315             |
| 10                                | 016439              | 150                | 200 | 230                                       | 329                            | 0.360             |
| 11                                | 013051              | 150                | 200 | 240                                       | 383                            | 0.430             |
| 12                                | 013052              | 150                | 200 | 250                                       | 457                            | 0.490             |
| 13                                | 013053              | 160                | 200 | 260                                       | 625                            | 0.576             |
| 15                                | 013054              | 165                | 210 | 290                                       | 697                            | 0.765             |

Table 3

| SIZES AND WEIGHTS OF STOUT POLES |                     |                    |     |                                           |                                |                   |
|----------------------------------|---------------------|--------------------|-----|-------------------------------------------|--------------------------------|-------------------|
| LENGTH                           | ITEM<br>CODE<br>Nos | DIAMETER AT<br>TOP |     | MIN. DIAMETER<br>AT 1.5m FROM<br>BUTT END | MAXIMUM<br>WEIGHT IMP.<br>RED. | NOMINAL<br>VOLUME |
|                                  |                     | MIN                | MAX |                                           |                                |                   |
|                                  |                     | mm                 | mm  | mm                                        | kg                             | m <sup>3</sup>    |
| 9                                |                     | 190                | 230 | 275                                       | 379                            | 0.460             |
| 10                               |                     | 190                | 230 | 285                                       | 430                            | 0.550             |
| 11                               |                     | 190                | 235 | 295                                       | 510                            | 0.653             |

**Imp Red - Imported European Redwood**

NB: The maximum tip diameter may be exceeded in exceptional circumstances by agreement with the SA.

**7.2 Measurement**

Minimum Diameters - The diameter shall be measured by callipering in 2 directions at right angles to each other, the average diameter then obtained shall be valid.

Maximum Diameters - The diameter shall be measured by callipering, the maximum diameter of the pole at that cross section shall be valid.

## 8 MARKING AND DRILLING

### 8.1 Marking

After final dressing each pole shall be marked in accordance with Drawing CN 351 by gouging or routing provided it can meet the specification in all respects. Poles treated with copper-oil or water-based-copper preservatives shall be marked accordingly, as detailed in Drawing CN 351 & Appendix A.

Within the limitations of the marking system in use, the depth of the 3 m markings shall be as close as possible to, but not exceeding the dimensions given on CN 351. (No exact minus tolerance is given owing to the difficulty of achieving consistently accurate results on the highly variable surface of a pole). The 3 m reference mark must be 3 m from the butt +50, -0 mm.

The 3 m reference mark shall be offset around the pole from the line of the drilling at the tip by 45° +/- 20°.

Alternative forms of marking will be considered providing they contain a minimum of the data above in the marking and are confirmed to have a service life in UK weather conditions of 40 years. Additionally any such marking must incorporate the batch number. Where alternative markings are proposed, they must incorporate the provision of the depth reference line shown in CN 351 and Appendix A of this document, provided by gouging or routing.

The butt of each pole shall be stamped to show the length and class of the pole as follows:

- Class to be marked by stamp min 20 mm in height, 'L' for Light Poles, 'M' for Medium Poles, S for Stout Poles, followed by:
- Length in metres to be stamped clearly in numerals, min 20 mm in height.
- Preservative type used.

### 8.2 Drilling

The pole shall be drilled at the tip in accordance with Drawing CN 206.

### 8.3 Charge Identification

The tip of the pole shall have a Latchbacher (or similar weatherproof label or disc) attached, to enable the charge number and supplier to be identified. The colour of the label must not be red.

## 9 KILN DRYING

Generally, kiln dried poles are not acceptable. Any request for approval to supply kiln dried poles must include validation of the techniques from an independent third party of recognised competence. Approval is at the discretion of the BT SA.

The validation submitted with any request for approval shall include details of the effects of the proposed drying regime on the poles in terms of strength, preservation and serviceability. The full range of the drying schedule variations shall have been assessed, and the validity of the defect monitoring system must be included.

### 9.1 Pre-warming of untreated timber

In locations where very cold conditions may prevail (i.e. Nordic countries) pre-warming of untreated poles immediately prior to treatment may be permitted, subject to approval in writing by the BT SA. The primary purpose of pre-warming is to raise the core temperature of the timber to facilitate a more successful treatment cycle, not to remove excess moisture.

Where pre warming is approved by the BT SA, a Maximum heating chamber temperature of 50°C shall be applied slowly over a period of up to 24 hours. This period may be exceeded in exceptional circumstances by agreement with the SA.

## 10 PRESERVATION

### 10.1 Pre-Treatment moisture content

The moisture content of the wood shall be determined in accordance with BS EN 13183 (all parts) and shall be appropriate for the preservative (as specified by the preservative manufacturer), treatment method and end use.

For poles treated with creosote: Preservation of a charge of poles shall not take place until the average moisture content of the charge of poles is reduced to not more than 28% and no individual pole in the charge has a moisture content greater than 30%.

For Poles treated with copper--oil-based, or water-based-copper preservatives:

Preservation of a charge of poles shall not take place until the moisture content of all individual poles is reduced to not more than 28%

For all preservative types: The moisture content of the poles shall be determined prior to treatment by the oven test method described in Appendix B of BS 1990 or by use of moisture meters, or by other methods agreed with the BT SA. When using moisture meters, any measurement over 25% shall be confirmed by the oven test method.

Records of moisture checks must be retained by the supplier for a minimum period of 7 years.

Where the sapwood band is greater in depth than the effective working depth of the moisture meter employed, then test methods agreed by BT SA shall be used to ensure that the sapwood not measured by meter is not above 30%. In this instance averaged moisture readings across the sapwood band are not acceptable.

Prior to treatment a 100% sample of poles shall be checked using a moisture meter at a position 1-2m from the butt. The minimum number of checks shall be as follows:

- Poles to be treated with Creosote – Check at two positions at 90° to each other.
- Poles to be treated with Copper preservatives – Check at three positions 120° apart

Batches of poles selected for preservation and checked for moisture level shall not be exposed to rain or any other form of precipitation until after preservation.

## **10.2 Preservatives**

Any preservative must be approved for use under the European Biocidal Products Regulations (BPR) or equivalent BPR in GB and shall conform to the performance requirements of Use Class 4 detailed in BS EN599-1.

### **10.2.1 Creosote**

Coal tar creosote used shall comply with EN13991-Grade B or C. Where Grade B is used, the Benzo(a)pyrene content must not exceed that of Grade C (50 parts per million by mass) and water extractable phenols (3 % by mass).

Requests for the use of alternative creosotes may be considered at the discretion of the SA. Full details of the creosote and its variation from EN13991-Type B or C must be supplied with any request.

Adequate measures shall be taken to ensure that the creosote used for preservation remains within specification until such time as it is incorporated into poles or removed from the treatment/storage cycle.

### **10.2.2 Copper type preservatives**

The preservative used shall be either:

Copper-oil-based and used as the sole method of preservation.

or

Water-based-copper preservatives. Used in conjunction with a separate hot oil barrier impregnated into the outer sapwood (e.g. RVP Repellent)

The use of water-based-copper preservatives as the sole method of preservation is currently not approved by BT.

The desired service life (DSL) is 40 Years, and for any preservative product used, there must be data from the preservative manufacturer available which supports this.

As per BS 8417:2024 -The supplier must also hold and follow safe relationship recommendations (provided by the preservative manufacturer) for achieving the DSL.

It is recognised that (unlike Creosote) copper-based preservatives are brand and product specific and that these may change over time. From the outset, the supplier shall provide to the SA details of the specific preservative/s to be used (including the relevant Material Safety Data Sheet) and receive approval for its use in writing from the SA. This requirement also applies where the supplier intends to switch from a currently approved preservative product to a new one.

## **10.3 Preservation Method**

### **10.3.1 Creosoted Poles**

The poles shall be pressure impregnated with creosote by the empty-cell (Reuping) process in accordance with BS 8417:2024 & BS 144:1997 to the standard detailed below.

It is accepted that wood poles are a naturally variable product and that the creosote retention of individual poles and batches will vary significantly.

The poles supplied shall be treated to BS 8417:2024 & BS144:1997 in such a manner as to produce a minimum net whole pole creosote retention of 115 kg/m<sup>3</sup> in imported European Redwood (*Pinus sylvestris*) with an average sapwood content of 66% measured at the butt. Full sapwood penetration is essential, as detailed in BS EN 351 Part 1 - 2007

The figure of 115 kg/m<sup>3</sup> may be adjusted for different species, batches or charges of poles based on the average or actual sapwood volume when this differs from the average quoted above, with the treatment process adjusted to aim for a net retention which is the equivalent of 174 kg/m<sup>3</sup> in the sapwood of the poles concerned.

Average values for sapwood volume shall be obtained by statistically valid methods.

Pressure rise times may be reduced from those indicated at the discretion of the supplier, but the time from initial air pressure rise above atmospheric to the commencement of filling shall not be less than 30 minutes (any holding time shall be at maximum pressure).

The period at full pressure shall be variable depending on the judgement of the supplier, but shall comply at all times with minimum pressure period defined in BS 144: 1997

Creosote temperature in the treatment cylinder shall conform to BS 144: 1997.

### **10.3.2 Poles treated with copper type preservatives.**

The poles shall be pressure impregnated with preservative using a suitable high-pressure treatment process.

It is accepted that wood poles are a naturally variable product and that the preservative retention of individual poles and batches will vary.

The poles supplied shall be treated to give full sapwood penetration of the preservative and the minimum retention appropriate to meet the recommendations for a desired service life of 40 years in accordance with BS8417:2024. Overall retention figures for individual preservatives may vary, but in any case, the retention requirement for a copper-oil or water-based-copper preservatives shall be equal to or above 3 kg/m<sup>3</sup> as copper in the analytical zone (penetration class NP5: full sapwood).

Treatment parameters such as pressure, vacuum, times and temperature shall be selected to achieve the penetration and retention requirements and be in accordance with the safe relationship recommendations of the preservative manufacturer. In particular, the following:

- Maximum temperature of heating media
- Maximum temperature of preservative
- Minimum temperature of preservative during pressure phase
- Max % water content

## 10.4 Measurement of Retention

**For Creosoted and for Copper-oil-based poles:** The whole pole retention shall be measured by weight to an accuracy of  $\pm 3\%$ . Selected poles, (minimum 6 per treatment charge) shall be identified to allow weighing before and after treatment. The weight difference in kilograms shall be divided by the cubic content of those poles and expressed in  $\text{kg}/\text{m}^3$ .

Creosoted retention shall:

- have a value of not less than **115**  $\text{kg}/\text{m}^3$  net (whole pole) charge average in imported European Redwood (*Pinus sylvestris*) with an average sapwood/heartwood ratio of 66%, or equivalent as defined above, when averaged over any 20 sequential batches.

and

- not be less than **107**  $\text{kg}/\text{m}^3$  net (whole pole) per charge average in imported European Redwood (*Pinus sylvestris*) with an average sapwood/heartwood ratio of 66%, or equivalent as defined above

Copper-oil-based retention shall:

- have a value of not less than **100**  $\text{kg}/\text{m}^3$  sapwood retention (66 $\text{kg}/\text{m}^3$  equivalent net whole pole retention) per charge average in imported European Redwood (*Pinus sylvestris*) with an average sapwood/heartwood ratio of 66%, or equivalent as defined above, when averaged over any 20 sequential batches. See table 4 for further details.

**For Poles treated with Water-based-copper preservatives:**

The actual copper content retention shall be measured / demonstrated using an X-Ray fluorescence analyser.

The retention shall:

- have a value of not less than that shown in Table 4 below, net per charge average in imported European Redwood (*Pinus sylvestris*) with an average sapwood/heartwood ratio of 66%, or equivalent as defined above.

Alternative volume methods of retention measurement are not acceptable but will be considered for acceptance on application to the BT SA.

All charge information and preservative retention records shall be kept for a minimum of 5 years.

**Table 4 – Retention quantities for Copper type preservatives**

| WOOD PRESERVATION PRODUCT                                                 | SAPWOOD RETENTION Kgs/m <sup>3</sup> | COPPER CONTENT SAPWOOD RETENTION Kgs/m <sup>3</sup> | TREATMENT TYPE ACRONYM |
|---------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------|------------------------|
| ARXADA TANASOTE S40                                                       | 100                                  | 3                                                   | TS                     |
| KOPPERS POLE PROTECT (OS-320B)                                            | 100                                  | 3                                                   | PP                     |
| RVP REPELLENT (*using Tanalith E 3475).<br>Step 1 - Cu-Water preservative | 34                                   | 3                                                   | RV                     |
| Step 2 - Hot oil application                                              | 70                                   | N/A                                                 |                        |

\*Where a different preservative to Tanalith E 3475 is proposed to be used, the specific manufacturers retentions figures for that product shall be shared / agreed with the BT SA.

## 10.5 Penetration

Regardless of the type of preservative used. All the sapwood shall be completely penetrated with preservative. Samples must be taken from every charge as per Table 5 below. The core samples shall be taken between 400 and 600mm from the butt of the pole.

Testing shall be carried out to a schedule agreed with the BT SA. The heartwood - sapwood boundary shall be defined by a chemical staining method. The samples shall be taken from poles both on the inside and outside of the charge. The sample cores shall be retained for a period of either 12 months, or until the next routine QA audit, whichever is longer. All holes from cores taken for testing must be plugged with creosote treated plugs, which are an interference fit into any holes made.

**Table 5 – Penetration core sampling**

| CHARGE SIZE | No OF SAMPLES TAKEN FROM CHARGE | MAXIMUM No OF SAMPLES WITH LESS THAN 100% PENETRATION IN THE SAPWOOD |
|-------------|---------------------------------|----------------------------------------------------------------------|
| 1 - 15      | 3                               | 0                                                                    |
| 16 - 25     | 5                               | 0                                                                    |
| 26 – 50     | 8                               | 0                                                                    |
| 51 – 90     | 13                              | 0                                                                    |
| 91 – 150    | 20                              | 0                                                                    |
| 151 – 280   | 32                              | 0                                                                    |

## 10.6 Storage and Pole Condition On and After Delivery to BT

Bleeding poles are not acceptable and shall not be supplied. A bleeding pole is one classified as WET (as defined within the glossary and the bleeding scale shown within table 6 of this specification).

Where Copper Oil and Creosote Poles are being transported together there shall be no cross contamination of creosote onto poles treated with copper type preservatives.

## 11 GLOSSARY OF TERMS

|                        |                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definitions</b>     | Any terms not defined below should be as defined in BS 6100: current issue and BS EN351-1:2007 and BS EN 599-1:2009.                                                                                                                                                                                                                                                                        |
| <b>Ambrosia Damage</b> | <p>Beetle damage characterised by round holes up to about <b>3 mm</b> in diameter; the tunnels are usually darkly stained (see Stain) not containing bore-dust (frass) and penetrating the timber mainly cross the grain.</p> <p>Ambrosia beetles are commonly called PINHOLE BORERS or SHOTHOLE BORERS, according to the size of the holes. They are active in unseasoned timber only.</p> |
| <b>Batch</b>           | Any group of poles.                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Bird Peck</b>       | A small hole or patch of distorted grain, sometimes associated with discolouration, attributed to the action of the birds.                                                                                                                                                                                                                                                                  |

|                         |                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bleeding</b>         | Exudation of preservative from the surface of a treated pole after the treatment process is complete. See bleeding scale f (Table 6) for further detail.                                                                                                                                                                                 |
| <b>Blue Stain</b>       | The most common form of sapstain producing a bluish discolouration.                                                                                                                                                                                                                                                                      |
| <b>Butt Sweeps</b>      | A deviation from straightness within 1.5 m of the butt.                                                                                                                                                                                                                                                                                  |
| <b>Charge</b>           | All those poles treated with creosote together in one treatment vessel at one time using the same process.                                                                                                                                                                                                                               |
| <b>Compression Wood</b> | Abnormal wood (reaction wood) formed typically on the lower sides of branches and of leaning or crooked trunks of softwood (coniferous) trees. Zones of compression wood are typically denser and darker than the surrounding tissue and have abnormally high longitudinal shrinkage tending to cause warping and splitting.             |
| <b>Rot (Decay)</b>      | Decomposition by fungi and other micro-organisms resulting in softening, progressive loss of strength and weight and often a change of texture and colour.                                                                                                                                                                               |
| <b>Growth Ring</b>      | A layer of wood produced in one growing season, seen as a ring in cross section.                                                                                                                                                                                                                                                         |
| <b>Grubhole</b>         | A worm hole usually more than 3 mm in diameter. Such holes are typically due to Bee hole borers, Carpenter ants, Jewel beetles, Longhorn beetles or Wood wasps. They may be round or oval in section, with or without fibrous bore-dust (frass). This term does not normally apply to damage by organisms that attack timber in service. |
| <b>Inbark</b>           | Bark in a flute or pocket or associated with a knot which has been partially or wholly enclosed by the growth of the tree; resin or gum may sometimes be present in the pocket.                                                                                                                                                          |
| <b>Knot</b>             | A portion of a branch enclosed in the wood by the natural growth of the tree.                                                                                                                                                                                                                                                            |
| <b>Knot Cluster</b>     | Set of knots close to each other and which can be considered as one defect. The grain of the pole is deviated around the set of knots and not around each separate knot.                                                                                                                                                                 |
| <b>Knot, Sound</b>      | A knot free from decay, solid across its face and at least as hard as the surrounding wood.                                                                                                                                                                                                                                              |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knot, Unsound</b>  | A knot not solid across its face or softer than the surrounding wood due to decay or other defects.                                                                                                                                                                                                                                                                                                           |
| <b>Pin Hole</b>       | A worm hole usually not more than 1.5 mm in diameter. Such holes are typically due to Ambrosia beetles. Usually they are darkly stained (seen Stain) and do not contain bore-dust (frass). This term does not normally apply to damage by organisms that attack timber in service.                                                                                                                            |
| <b>Punk</b>           | A wound or organic damage pocket in the wood where decay or <b>Punk Knot</b> infection has occurred, and growth has failed to re-establish. A funnel-like cavity often filled or partially filled with non-fibrous matter. Normally caused by the fruiting body of <i>Trametes pini</i> and is indicative of possible significant heart rot.                                                                  |
| <b>Rate of Growth</b> | <p>The growth expressed as the number of growth rings per 25 mm measured radially on the end grain of a log or piece of converted timber.</p> <p>Rate of growth shall be measured at one end of the piece of timber and shall be expressed as the average number of growth rings intersected by a radial line 25 mm long. Measurements shall be made over a radial line as long as possible, up to 75 mm.</p> |
| <b>Rindgall</b>       | A surface wound that has been enclosed by the growth of the tree.                                                                                                                                                                                                                                                                                                                                             |
| <b>Sapstain</b>       | A discolouration of timber resulting from the growth of certain fungi that derive their nourishment from the cell contents but do not cause decomposition of the timber. It is principally confined to sapwood.                                                                                                                                                                                               |
| <b>Sapwood</b>        | The outer layers of wood which, in the growing tree, contained living cells and reserve materials (e.g. starch); generally lighter in colour than heartwood though not always clearly differentiated.                                                                                                                                                                                                         |
| <b>Shake</b>          | A separation of the fibres along the grain due to stresses developing in the standing tree, or in felling or in seasoning.                                                                                                                                                                                                                                                                                    |
| <b>Shake, Heart</b>   | A radial shake originating at the heart.                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Shake, Star</b>    | A number of heart shakes more or less in the form of a star having 5 or more points.                                                                                                                                                                                                                                                                                                                          |
| <b>Shake, Ring</b>    | A shake following a growth ring.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Short Crook</b>    | A localised deviation from straightness within any pole section of 1.5 m or less.                                                                                                                                                                                                                                                                                                                             |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shothole</b>         | A worm hole more than 1.5 mm and not more than 3 mm in diameter. Such holes may be due to Ambrosia beetles in which case they are usually darkly stained (see Stain) and not containing bore-dust (frass), or to Powder-post beetles, in which case they are unstained (see Stain) and may contain bore-dust (frass). This term does not usually apply to damage by organisms that attack timber in service. |
| <b>Spiral Grain</b>     | Grain which follows a spiral course in one direction round the stem.                                                                                                                                                                                                                                                                                                                                         |
| <b>Turpentine Scars</b> | <p>Acid Face. An area in the lower portion of a Southern Pine pole where bark hack removal with acid applied has caused resin to flow.</p> <p>Cat Face. A depression in the surface of a Southern Pine pole resulting from a wood hack into the sapwood, where healing has not re-established the normal cross section of the pole.</p>                                                                      |
| <b>Worm</b>             | Any wood-boring organism (chiefly insects but including marine borers) that attacks wood producing holes or tunnels or similar damage.                                                                                                                                                                                                                                                                       |
| <b>Wood wasp Damage</b> | Damage characterised by round holes up to 6 mm in diameter; the tunnels are densely packed with bore-dust (frass). Found chiefly in softwood.                                                                                                                                                                                                                                                                |
| <b>Worm Hole</b>        | A hole or tunnel, irrespective of size, caused by worm.                                                                                                                                                                                                                                                                                                                                                      |

## 12 BLEEDING SCALE

Bleeding shall be classified according to the following scale where for the purpose of test, 'hand' shall mean a hand covered with a light-coloured cotton glove, rested on the pole, and 'residue and film' shall be preservative which has soaked into the fibres of the glove.

**Table 6 – Bleeding scale**

| <b>Pole appearance</b>                                                                                                    | <b>Test outcome</b>                 | <b>Pole status</b> |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------|
| Surface is dry other than minimal wetness around knots.<br>No evidence of film or surface deposits elsewhere on the pole. | No residue on hands away from knots | DRY                |
| Pole has areas other than that around knots which appear wet.<br>Film or surface deposits of preservative evident.        | Residue on hands away from knots    | WET                |

## 13 PRODUCT APPROVAL

Product Approval shall be in accordance with BT Specification CW 2000. In addition to the requirements of CW2000, pole suppliers will be required to supply a comprehensive data pack, which provides all the necessary detail and supporting evidence, on how specification LN435 is met and complied with.

As a minimum, this pack will include: -

- Documentary evidence to show compliance with the requirements on Sourcing, Forest Management, Felling and Storage of poles.
- Detail of independent testing.
- Detail of the Inspection regime, how poles are checked for defects and dimensions
- Detail of the pole dressing and fabrication processes, and readiness for preservative treatment.
- Full details of the preservation process, both creosote and alternatives, and how the penetration and retention are both verified and recorded
- Company procedures for dealing with non-conforming products, e.g. identification, retreatment, re-marking etc.
- Staff training and accreditation
- QA procedures, including detail of recognised quality procedures, e.g. ISO 9001
- Storage and transport of completed poles

Final contents of the data pack will be subject to approval and acceptance by the SA

## 14 REFERENCES

This specification makes reference to, or should be used in conjunction with the following documents:-

|                  |                             |
|------------------|-----------------------------|
| British Standard | BS 1990: Part 1:<br>1984    |
| British Standard | BS EN 14229: 2010           |
| British Standard | BS 144: 1997                |
| British Standard | BS 6100: current<br>version |
| British Standard | BS EN 351-1:2007            |
| British Standard | BS EN 599-1:2009            |
| British Standard | BS8417:2024: 2011           |
| BT Specification | GS 13                       |
| BT Specification | CW 2000                     |

BT Drawing CN 206  
BT Drawing CN 351

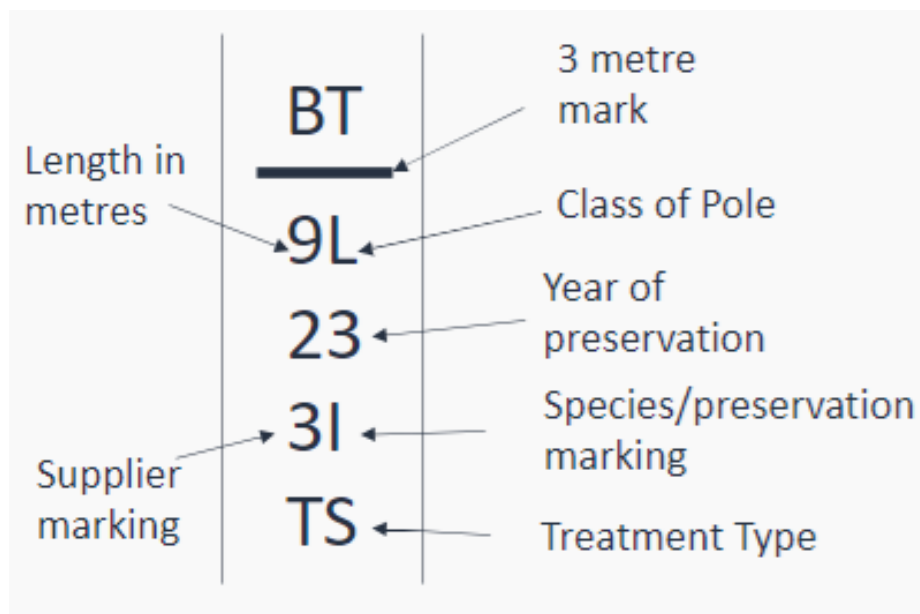
## 15 HISTORY

| ISSUE           | DATE        | CHANGES                                                                                                                              |
|-----------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------|
| LN 435 Draft    | 05.11.86    | Amalgamation of LN 23 and LN 147.                                                                                                    |
| LN 435          | 03.12.88    | Corsican, Larch, Douglas removed Straightness tip to butt measurement intro.                                                         |
| LN 435A         | 05.10.89    | Insect holes, mechanical damage parameters.                                                                                          |
| LN 435B         | 01.04.90    | Trimming, butt flutes, shakes, allowances.                                                                                           |
| LN 435C Draft 1 | 20.07.90    | Rewritten. Marking tip, rindgall draw-knifing.                                                                                       |
| LN435C Draft 2  | 29.08.90    | Rewritten. Soft knot test spiral grain assay test.                                                                                   |
| LN 435C         | 22.11.90    | Implementation of Draft 2 – minor changes to layout and general notes only.                                                          |
| LN 435D         | Not Issued  | As LN 435C with the addition of notes.                                                                                               |
| LN 435E         | 02.11.93    | Preservation rewritten, significant changes to defects, some changes to other sections.                                              |
| LN 435F         | 04.02.94    | Changes to marking, moisture, size/weight, etc. Environmental clauses added.                                                         |
| LN 435G         | 08.05.98    | Changes to marking, sources, preservation kiln drying, addition of CW 200 requirement.                                               |
| LN 435H         | Sep 2012    | Changes to species, treatments, kiln drying, defect assessments. Addition of stout poles. Updates to BS & BS EN references           |
| LN435J          | Oct 2012    | Correcting incorrect years for quoted BS documents                                                                                   |
| LN435K          | July 2013   | Removal of home grown option, and concession on butt flutes                                                                          |
| LN435L          | Aug 2014    | Change to storage time for core samples for penetration test & stipulation for retaining of charge / preservative retention records. |
| LN435M          | Feb 2015    | Changes to permitted moisture levels for alternative treatment poles                                                                 |
| LN435(N)        | August-2024 | Major changes to section 9 and 10 to incorporate Copper treated poles. Also, minor changes and updates throughout document.          |

## 16 SPECIFICATION AUTHORITY

**Wesley Grantham**  
**Openreach Overhead Policy & Standards Manager**  
[Wesley.grantham@openreach.co.uk](mailto:Wesley.grantham@openreach.co.uk)

Example of a 3-Metre mark with Alternative treatment, treatment type acronyms can be found within Table 4.



**END OF SPECIFICATION**